Reducing Biofilms on Surgical Mesh Through Controlled Electrochemical Current Treatment as an Alternative Approach to Combating Antimicrobial Resistance

Introduction

Surgical biomaterials such as polypropylene mesh are widely used in clinical procedures. Approximately **15% of hospital waste is hazardous**, with a signifcant portion containing infectious agents capable of spreading disease and antimicrobial resistance (World Health Organization, 2018). Bacteria that attach to polymeric surfaces frequently form biofilms, structured microbial communities encased in a protective extracellular matrix.

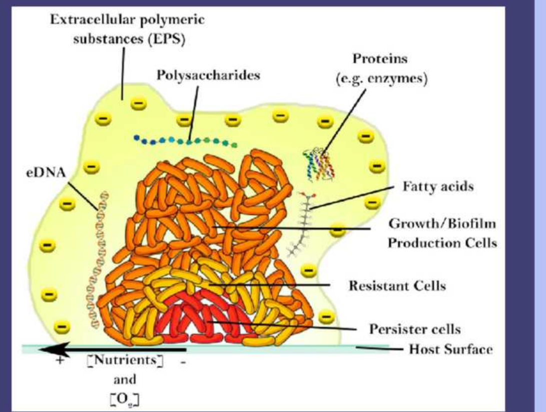


Figure 1. Biofilm Structure (Benetton, 2007).

If not effectively neutralized prior to disposal, contaminated mesh can contribute to **pathogen spread** within healthcare environments and waste management systems. Traditional sterilization methods, including chemical disinfectants, antibiotics, and autoclaving, are often impractical for hospital wastewater systems because of high cost, environmental toxicity, and incomplete biofilm removal.

Figure 2. Surgical Mesh with Bacteria Biofilm (Michigan Hernia Study, 2023).

Electrochemical current has emerged as a potential non-antibiotic approach for **disrupting membranes** and destabilizing biofilm **structure.** However, its application as a decontamination strategy for used surgical mesh remains underexplored.

Purpose

Evaluating whether controlled electrical current can **reduce pathogen-associated biofilms** on surgical mesh prior to disposal. Targeting resistant, real-world polymicrobial biofilms at the disposal stage, rather than single-species models, better reduces pathogen persistence, **antimicrobial resistance risk**, and **environmental contamination**.

Methods

PHASE 1 — Biofilm Reduction via Electrochemical Treatment

Culture Preparation → **Electrochemical Treatment** → **Plating & Quantification** → **Data Analysis**

Culture Preparation

- Prepared mono- and polymicrobial cultures
  - *E. coli K-12*
  - *P. fluorescens*
  - *S. epidermidis*
- Grown in TSB for 48 hrs to form biofilms

Electrochemical Treatment

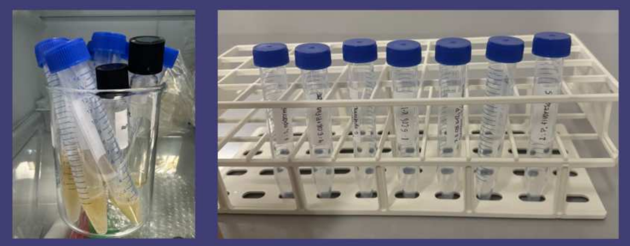
- Selected Conditions:
  - Control - 0 kV
  - Low intensity - 1.0 kV, 1.0 ms pulse
  - High intensity - 2.5 kV, 1.0 ms pulse
- Applied current using cuvettes

Plating & Quantification

- Agar plating of treated cultures
- Incubation period length: approximately 120 hrs
- Colony counts (CFU) recorded for quantification

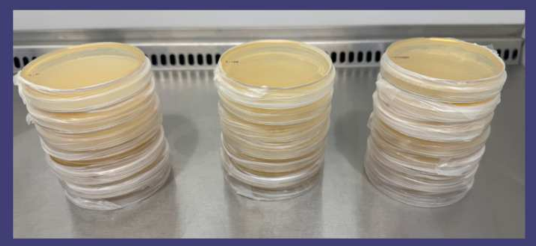
Data Analysis

- Voltage vs. Bacterial Survival:
  - 1.0 kV → 16% reduction
  - 2.5 kV → 20% reduction
- Mixed cultures have the highest resistance over single-species



Figures 3 and 4. Liquid Cultured Species (Photos taken by Finalist).

Figures 5 and 6. Electrical Treatment (Photos taken by Finalist).



Figures 7. Plating All Cultures (Photo taken by Finalist).

Figures 8-10. Culture #7 Growth Results for (Photos taken by Finalist).

PHASE 2 — Chamber Construction

System Configuration

- Acrylic box (sealed system)
- Graphite electrode plates

Electrode Holder

- 3D Designed on Inventor
- Made to be compatable with electrodes

Surgical Mesh Holder

- 3D Designed piece and printed using PLA filament
- Mesh placement rack centered between electrodes

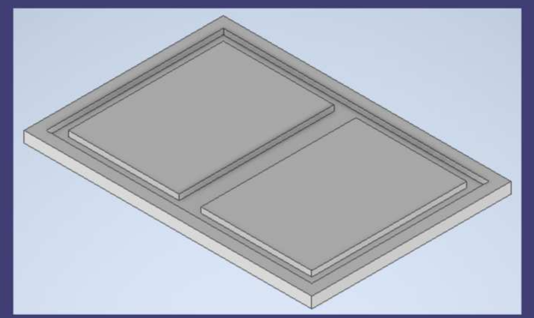


Figure 11. Chamber Base (Image created by Finalist using Inventor)

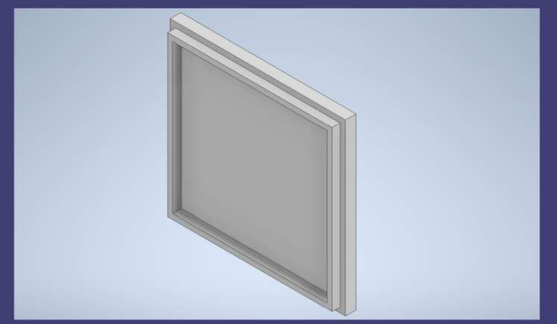


Figure 12. Mesh Holder (Image created by Finalist using Inventor)

Figure 13. Fully constructed Chamber (Photo taken by Finalist).

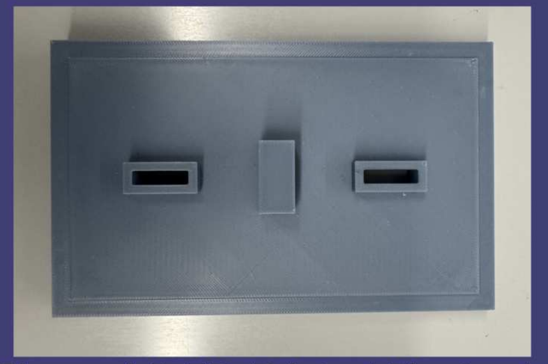


Figure 14. 3D-Printed Electrode Holder (Photo taken by Finalist).

PHASE 3 — Application Testing

1 Simulated Conditions

- Culture bacterial biofilm on surgical mesh
- Used 3 organisms:
  - *E. coli K-12*
  - *P. fluorescens*
  - *S. epidermidis*
- Used Electrolyte Solution (NaCl)

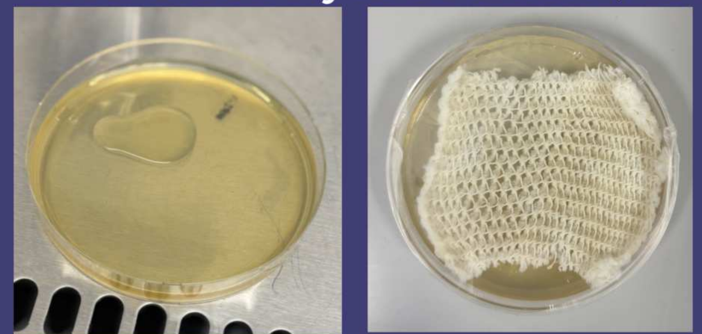


Figure 15 and 16. Culturing Bacteria on Surgical Mesh (Photos taken by Finalist).

2 Treatment of Mesh

- 3 groups:
  - control (no current)
  - 20 V, 180 s pulse
  - 25 V, 180 s pulse
- Controlled current applied
- DC Power Supply Used to provide current

Figure 17. Use of DC Power Supply to Administer Treatment (Photo taken by Finalist).

3 Incubation & Analysis

- Treated groups were plated on agar plates; 72-hr period
- CFU count was recorded for all three conditions
- Microscopic imaging to view bacterial formation after incubation

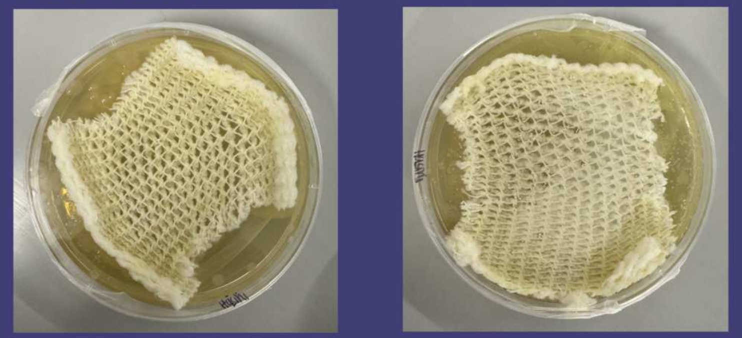


Figure 18 and 19. Post Treatment Cultures (left-no pulse, right-25 V) (Photos taken by Finalist).

Results

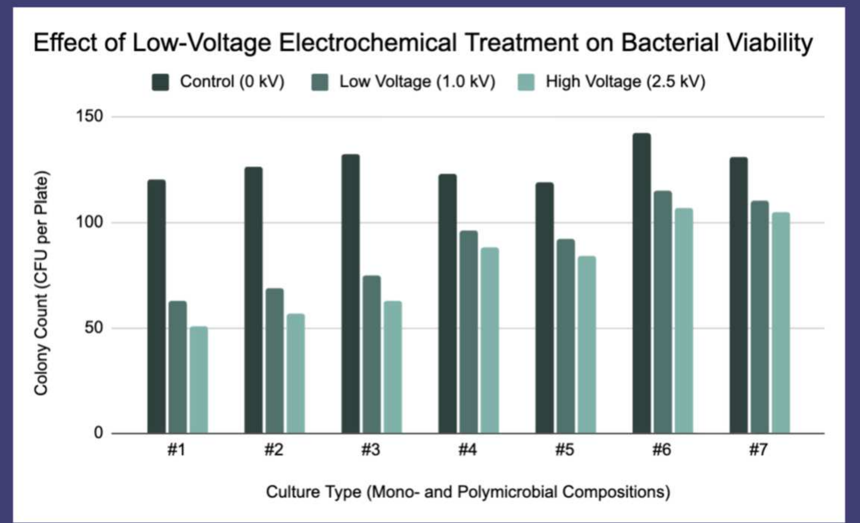


Figure 20. Phase 1 Results. Increasing applied voltage corresponded with decreases in colony-forming units (CFUs), with mixed-species biofilms indicating greater resistance than single-species (Graph created by Finalist using Google Sheets).

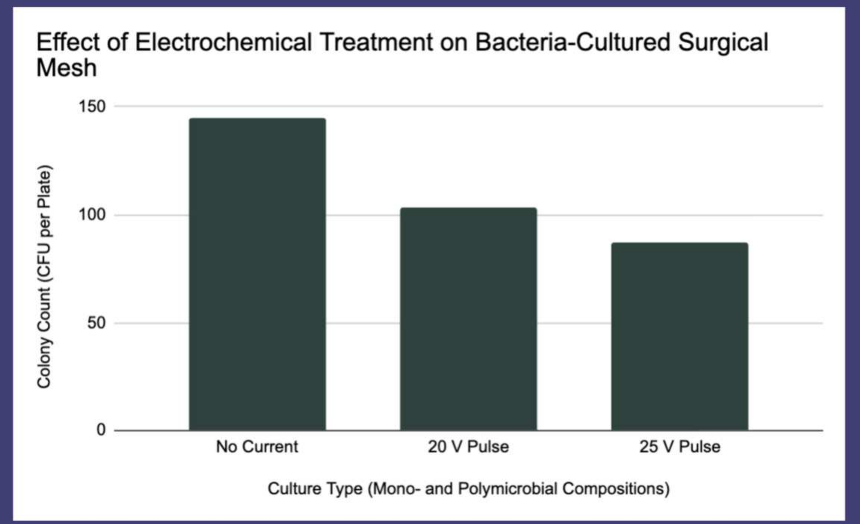


Figure 21. Phase 3 Results. Colony-forming unit (CFU) counts recovered from biofilm-coated surgical mesh after exposure to controlled electrical pulses within a sealed electrochemical chamber (Graph created by Finalist using Google Sheets).

Conclusions

High-voltage pulsed treatment caused a **voltage-dependent reduction** in bacterial viability, with 25 V producing the greatest CFU decrease in biofilms. The **sealed electrochemical chamber** improved treatment consistency, and although polymicrobial systems were more resilient, optimized parameters achieved significant growth reduction.

Pulsed electrochemical treatment demonstrates strong potential as a scalable, non-antibiotic strategy for reducing biofilm-associated pathogens on contaminated surgical materials prior to disposal, with future research focused on optimizing electrical parameters and evaluating **integration** into hospital wastewater and medical waste management systems.

Citations

Kim, Y. W., Subramanian, S., Gerasopoulos, K., Ben-Yoav, H., Wu, H., Quan, D., Carter, K., Meyer, M. T., Bentley, W. E., & Ghodssi, R. (2015). Effect of electrical energy on the efficacy of biofilm treatment using the bioelectric effect. NPJ Biofilms and Microbiomes, 1(1). https://doi.org/10.1038/npjbiofilms.2015.16

Landa, G., Clarhaut, J., Buyck, J., Mendoza, G., Arruebo, M., & Tewes, F. (2024). Impact of mixed Staphylococcus aureus–Pseudomonas aeruginosa biofilm on susceptibility to antimicrobial treatments in a 3D in vitro model. Scientific Reports, 14, 27877. https://doi.org/10.1038/s41598-024-79573-y

World Health Organization. (2018). Safe management of wastes from health-care activities: A summary. WHO. https://apps.who.int/iris/handle/10665/25949